

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

Anonymous authors
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ABSTRACT

Diffusion probabilistic models (DPMs) have achieved substantial gains on generative tasks such as image and video synthesis. Nevertheless, their sampled outputs frequently contain *hallucinations*: instances that are inconsistent with the factual structure of the real world, for example a spurious duplicate cup rendered floating beside another cup. Although such artifacts are pervasive, the absence of a tractable methodology for systematically quantifying them impedes principled progress on the problem and leaves unspecified the design space for next-generation generative models subject to factual constraints. This work addresses that gap by isolating a well-scoped subclass of hallucination, which we term **counting hallucination**: the generation of an incorrect cardinality of instances or of structured objects (e.g., a hand image exhibiting six fingers) even though no such configuration is present in the training distribution. To support controlled study, we construct a dataset suite, **CountHalluSet**, in which the admissible object counts are specified exactly; the suite comprises ToyShape, SimObject, and RealHand. On these datasets we formalize a standardized evaluation protocol for quantifying counting hallucinations, and we conduct a controlled analysis of how the sampling configuration of DPMs, specifically the solver type, the order of the ODE solver, the number of sampling steps, and the initial noise, governs the incidence of counting hallucination. We further quantify the statistical association between counting hallucination and standard evaluation metrics such as the Fréchet Inception Distance (FID), and demonstrate that widely adopted perceptual metrics do not reliably track counting hallucinations.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), hereafter diffusion models, constitute a class of generative models that have attained strong sample quality across a broad range of synthesis tasks, including text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Notwithstanding these advances, diffusion models remain unable to faithfully reproduce the factual regularities of their training distribution, and consequently emit outputs that are inconsistent with real-world knowledge. A representative failure is the synthesis of objects with an incorrect number of spatially intricate components, such as supernumerary human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

A growing body of work seeks to characterize and taxonomize the recurrent failure modes of text-guided diffusion models. Specifically, Borji (2023) provide a systematic qualitative account of the deficiencies exhibited by text-to-image models. Liu et al. (2024) formulate an adversarial search procedure that surfaces undesirable behaviors and failure cases in text-to-image generation, and Wu et al. (2023) adopt an iterative procedure to detect and attenuate spurious correlations.

More recently, *hallucination* has been identified as a distinct failure mode of diffusion models and has attracted increasing attention (Aithal et al., 2024). Prompt-conditioned failure modes admit a direct evaluation via the alignment between the conditioning input and the generated output (e.g., the CLIP score (Hessel et al., 2021) and the VQA score (Huang et al., 2025; Zarei et al., 2024)), and overt failures such as severe distortion, blur, or artifacts are readily detected; hallucinations, by contrast, are typically visually plausible, arise independently of the conditioning signal, and manifest

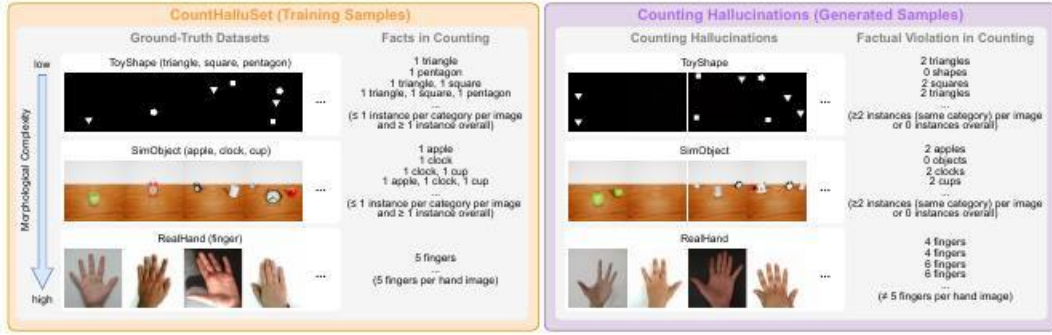


Figure 1: Illustrative comparison between the proposed **CountHalluSet** training data (left) and the **counting hallucinations** produced by a model trained on them (right). Each dataset is associated with an explicit counting criterion over its object categories. In ToyShape and SimObject, an admissible image contains at most one instance per category and at least one instance in total, whereas in RealHand an admissible image contains exactly five fingers. A counting hallucination is a generated image that violates the dataset’s counting criterion, e.g., two apples in a single image, or an image with a well-formed background but no objects at all, neither of which occurs in the SimObject training distribution.

as subtle violations of factual consistency with respect to the training distribution, for instance incorrect object counts, geometries that are perceptually plausible yet physically infeasible, or other violations of physical constraints. Resolving this fine-grained boundary between valid samples and hallucinations is a central difficulty for current systems, which detect such violations only unreliably. As a concrete illustration, even state-of-the-art large vision-language models such as GPT-4o fail on ostensibly elementary tasks, e.g., accurately enumerating the fingers in an unambiguous hand image. Consequently, a standardized and precisely specified protocol for reliably quantifying hallucinated samples is a prerequisite for principled progress: it establishes the basis for designing next-generation generative models under strict factual-consistency constraints, a capability that is essential to their intended role as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among the several attributes along which hallucinations manifest, namely count, color, spatial relationship, geometry, and physical realism, counting errors are especially amenable to systematic study, since they admit both unambiguous detection and exact quantification. This motivates our focus on a specific subclass of hallucination that we designate **counting hallucination**. *In this formulation, the admissible object counts of the training distribution are fixed a priori. A generated image is therefore labeled hallucinated precisely when its object counts fall outside this predetermined set.* For instance, an image depicting a hand with six fingers is a counting hallucination because such a configuration has zero support in the RealHand dataset, as shown in the last row of Fig. 1. To enable a controlled investigation of this phenomenon, we assemble the CountHalluSet suite of three datasets that span an increasing range of morphological complexity of the countable objects: ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger). Figure 1 presents representative samples from each dataset together with their corresponding counting hallucinations.

The central component of our quantification protocol is the accurate detection and enumeration of objects, for which we instantiate a dedicated counting model per dataset. Given these datasets and counting models, we quantify how the denoising configuration of the diffusion model, parameterized by the number of sampling steps, the initial noise, the ODE solver order, and the solver type, controls the counting hallucination rate. We additionally estimate the correlation between the counting hallucination rate and perceptual metrics, specifically the Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021).

Our principal findings and contributions are as follows:

- Numerical refinements that are standard in diffusion sampling, e.g., increasing the number of sampling steps from 25 to 100, fail to reduce counting hallucinations in human-hand generation, despite being effective on the simpler synthetic datasets.

- The counting hallucination rate is only weakly correlated with perceptual metrics, whereas the non-counting failure rate is strongly correlated with the same metrics.
- Motivated by these observations, we introduce a training paradigm, termed the joint-diffusion model, that substantially reduces both counting hallucinations and non-counting failures on the RealHand dataset.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Paralleling the extensive study of hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), hallucination in unconditional diffusion models has become an active subject of investigation. In particular, Aithal et al. (2024) advance a qualitative hypothesis attributing hallucinated samples to interpolation between distinct modes of the data distribution. This hypothesis notwithstanding, a systematic quantitative characterization of hallucination in diffusion models is still absent.

This gap motivates us to develop the machinery for accurate detection and quantification of hallucinations in diffusion-generated outputs. Such an analysis both sharpens the understanding of these undesirable behaviors and makes explicit how the magnitude of denoising error, e.g., variation in the number of sampling steps or in the ODE solver order, modulates the severity of hallucination.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

The Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and the CLIP score (Hessel et al., 2021) are two widely adopted metrics for evaluating text-guided image generation. Evaluation procedures that are grounded in language models are, however, fundamentally limited for assessing hallucinated images. One such limitation is quantified by the statistical lower bound on the rate at which pretrained language models hallucinate certain fact types, which in turn upper-bounds the detection accuracy attainable by language models alone (Kalai & Vempala, 2024). A further limitation is the persistent accuracy gap between human annotators and LLMs on the classification of hallucinated textual content (Lin et al., 2022; Li et al., 2023).

In the image-generation setting, several quantitative metrics are in common use, including FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021). Each of these metrics is computed from feature embeddings produced by a pretrained Inception-v3 model (Szegedy et al., 2016). Whether such metrics are sensitive to the *severity of hallucination* in a generated dataset, however, remains an open question, which we address in this work through the lens of counting hallucinations.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Let the observed data be distributed as $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$. Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (equivalently, a *diffusion process*) that progressively corrupts the data by injecting Gaussian noise over T discrete timesteps. At each timestep t , the conditional marginal of the noisy sample $\mathbf{x}_t$ given the clean data $\mathbf{x}_0$ is Gaussian:

$$q_{0:t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_i\}_{i=1}^T$ denotes a predefined, timestep-dependent noise schedule, typically instantiated as a linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021) schedule. The coefficient $\bar{\alpha}_t$ parameterizes the signal-to-noise ratio at step t , with the boundary conditions $\bar{\alpha}_0 = 1$ (noiseless) and $\bar{\alpha}_T \approx 0$ (approximately pure noise). Applying the reparameterization to Eq. (1) yields the equivalent sampling form:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

In the limit of large T , the terminal marginal $q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ converges to the standard Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$. In standard DPM implementations (Ho et al., 2020; Nichol & Dhariwal, 2021), T is set to 1000.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (equivalently, the *denoising process*) of a DPM iteratively reconstructs the data distribution starting from the diffused terminal state $\mathbf{x}_T$. Concretely, a DPM predicts the injected noise ϵ from the noisy input $\mathbf{x}_t$ via a neural network $\epsilon_\theta(\mathbf{x}_t, t)$, referred to as the *noise prediction model*. The parameters θ are estimated by minimizing the denoising objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a per-timestep weighting function.

Ancestral Sampling. Given the terminal noise $\mathbf{x}_T$, samples from the data distribution $q_0(\mathbf{x}_0)$ are drawn by integrating the reverse process over the same T steps used in the forward process, a procedure termed *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1-\beta_t}}(\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

in which the sole intractable term is $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$, known as the *score function* (Song et al., 2022).

In practice, the DPM approximates the scaled score function $-\sqrt{(1-\bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ by the noise prediction model, from which the reverse transition in Eq. (3) can be expressed equivalently as:

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon, \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. The ancestral sampling transition in Eq. (4) is stochastic (Kloeden et al., 1992); it admits an associated deterministic counterpart whose marginals coincide, expressible as an ordinary differential equation (ODE), the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau)\mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2}\beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the exact trajectory that satisfies this ODE under the true score function $h(\mathbf{x}(\tau), \tau)$, initialized at $\mathbf{x}(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}(T))$. Discretize the interval as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, with $\tau_N = 0$. The variation-of-constants formula gives the exact one-step update:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator associated with the linear part $\frac{dz}{d\tau} = f(\tau)z$. The step size $\Delta t > 0$ controls the discretization granularity; because ODE-based solvers admit larger step sizes than ancestral sampling, they reduce the number of function evaluations, and the total step count is $T/\Delta t$. As the linear term $\mathcal{G}(t_2, t_1)$ admits a closed-form expression, accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) concentrate their numerical budget, e.g., higher solver orders and additional sampling steps, on approximating the nonlinear integral involving h . A more detailed theoretical treatment is deferred to Appendix A.1 through A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section specifies the methodology and datasets used to quantify counting hallucinations. We first formalize the counting criterion and the induced notion of counting hallucination, and then describe the datasets comprising the CountHalluSet suite. For each dataset we state its counting criterion, the corresponding set of counting hallucinations, and the standardized protocol adopted for their quantification.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We now give a general definition of the counting criterion associated with a training dataset and characterize counting hallucinations as violations of the corresponding constraints.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ denote a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, let $N_c(\mathbf{x})$ be the number of objects of category c in $\mathbf{x}$. The counting criterion constrains $N_c(\mathbf{x})$ to a predefined admissible set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), i.e., $N_c(\mathbf{x}) \in \mathcal{S}_c$. We further impose that every valid reference sample contains at least one object, i.e., $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C denotes the set of categories under consideration.

Counting hallucinations as deviations from the counting criteria. A generated sample $\hat{\mathbf{x}}$ is said to exhibit a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ if and only if the following two conditions are jointly satisfied:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ must be of sufficient visual quality that its background and constituent objects are unambiguously discernible. This precondition decouples counting hallucinations from other salient failure modes, such as severe degradation. We encode it as a binary *counting-ready indicator* $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, where 1 signifies that the image quality suffices for counting and 0 otherwise. In practice, $\mathbb{I}_{\text{CRI}}$ is realized by human annotation, by a classifier calibrated against human annotations, or by an equivalent automated criterion.
2. **Violation of counting facts.** A violation of the counting criterion occurs when there exists a category c with $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or when the image is object-free (i.e., $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$), which is inadmissible since every sample must contain at least one object. In practice, object counts in generated images are predicted by a counting model, whose outputs serve as evidence for such violations.

Accordingly, $\hat{\mathbf{x}}$ is labeled a counting hallucination (CH) precisely when its quality is sufficient for counting ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and its predicted counts violate the criterion ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

The CountHalluSet suite comprises three datasets for the study of counting hallucinations: ToyShape, SimObject, and RealHand. Representative samples from each dataset, together with their associated counting hallucinations, are shown in Fig. A1 and A2. Additional summary statistics are reported in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. The ToyShape dataset consists of 30,000 images containing three geometric shape categories: *triangle*, *square*, and *pentagon*. All shapes are white, occupy an equal area of 120 pixels, and are placed without mutual overlap within any image. The counting criterion admits at most one instance per shape category and requires at least one shape in total, as depicted in the top-left of Fig. 1. Formally, $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$, and for each category $c \in C$ we set $N_c(\mathbf{x}) \in \mathcal{S}_c = \{0, 1\}$ subject to $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. Under the criterion above, any image generated by a model trained on ToyShape that contains two or more instances of a single category constitutes a counting hallucination, e.g., the image with two pentagons in the top-right of Fig. 1. Empty images, i.e., those containing only the black background and no shape, are likewise classified as counting hallucinations. Representative examples are provided in Appendix A.5.

Counting model used in ToyShape. We employ a counting model to detect counting violations, omitting the counting-ready indicator because the simplicity of this dataset renders severely degraded images negligibly rare. Concretely, we fine-tune a ResNet-50 (He et al., 2016) on over 400k ToyShape images in which each category appears 0–3 times per sample (i.e., $\mathcal{S}_c = \{0, 1, 2, 3\}$), so as to cover

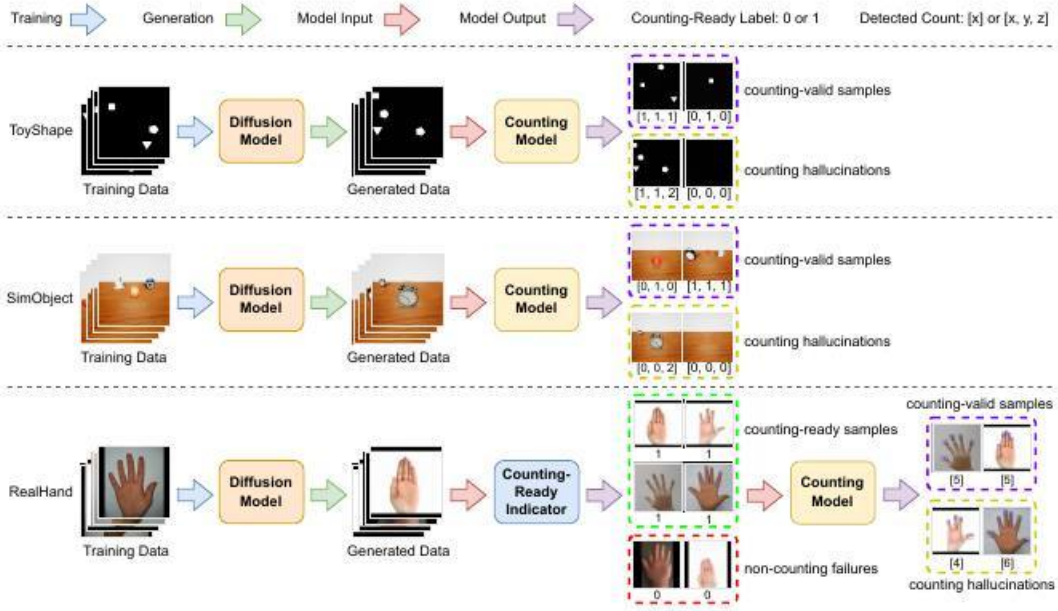


Figure 2: Evaluation pipeline of the CountHalluSet suite. For ToyShape and SimObject, the counting model is applied directly to every generated image. For RealHand, a counting-ready indicator (a binary classifier aligned with human judgments) is inserted upstream to separate counting hallucinations from non-counting failures, e.g., severely deformed fingers. Count labels are defined per dataset: ToyShape over *triangle*, *square*, *pentagon*; SimObject over *mug*, *apple*, *clock*; and RealHand over *finger*.

the full range of plausible configurations, including both counting-valid samples and counting hallucinations, and thereby enable accurate enumeration. The resulting model attains over 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. Given a diffusion model trained on ToyShape, we sample a set of images whose cardinality matches the training set (30,000 images). The counting model then predicts the per-category object counts for each generated image. Finally, images whose counts violate the criterion are flagged as counting hallucinations, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. The SimObject dataset contains 30,000 rendered images of everyday objects drawn from three categories, *mug*, *apple*, and *clock*, each with 10 distinct intra-class variations. Objects are randomly positioned on a wooden tabletop under a fixed lighting configuration with at most minimal mutual occlusion, as shown in the middle-left of Fig. 1. The images are synthesized in Unreal Engine 5 (Epic Games, 2024), which provides photorealistic rendering with faithful simulation of object geometry, material properties, lighting, and reflections. The counting criterion mirrors that of ToyShape (see Sec. 4.2.1): formally, $C = \{mug, apple, clock\}$, and for each category $c \in C$ we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$ subject to $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Owing to the equivalence of their counting criteria, the definition of counting hallucinations, the construction of the counting model, and the quantification protocol are identical to those of ToyShape, as shown in Fig. 2.

4.2.3 REALHAND

Dataset description. To study counting hallucinations in a real-world regime, we construct the RealHand dataset of 5,050 human-hand images aggregated from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². We pool the raw images from these sources and discard those exhibiting

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

blur or artifacts. Each retained image depicts a five-fingered human hand in either the dorsal (back) or palmar (front) view, under heterogeneous lighting conditions and backgrounds (e.g., white surfaces, blackboards, walls), as shown in the lower-left of Fig. 1. The counting criterion fixes exactly five fingers per image: $C = \{finger\}$ with $N_c(\mathbf{x}) \in S_c = 5$ for each $c \in C$, hence $\sum_{c \in C} N_c(\mathbf{x}) = 5$.

Counting hallucinations and non-counting failures. Realistic hand synthesis is susceptible to failures beyond counting hallucinations (e.g., extra or missing fingers), including blur, severely deformed fingers, and visual artifacts. We therefore introduce a counting-ready indicator, instantiated as a binary classifier, to isolate such cases. Examples of counting hallucinations and non-counting failures are provided in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To decouple counting hallucinations from the aforementioned failure modes, we fine-tune a MaxViT (Tu et al., 2022) classifier on 2.5k annotated generated images as the counting-ready indicator aligned with human judgments. For the counting model, we fine-tune a YOLO-12 (Tian et al., 2025) detector on 2k fingertip-annotated generated images. Evaluated by three annotators on 500 randomly sampled images, the indicator and counting model attain over 96% and 99% accuracy, respectively. Representative YOLO detection outputs are provided in Appendix A.6.

Quantification of counting hallucinations. In contrast to ToyShape and SimObject, where every generated image is evaluated directly because severe failures (e.g., distorted or unrecognizable objects) are rare, RealHand requires an additional filtering stage. Concretely, we first apply the counting-ready indicator to retain only images of sufficient visual quality, and subsequently apply the counting model to predict the finger count on the retained subset, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. For ToyShape we train a DDPM (Ho et al., 2020) from scratch; for SimObject and RealHand we fine-tune latent diffusion models (LDMs) (Rombach et al., 2022) pretrained on CelebA-HQ (Karras et al., 2018). The number of diffusion steps is fixed at $T = 1,000$ under a standard linear noise schedule. Input images are resized to 128×128 for the DDPM and to 256×256 for the LDMs. Optimization uses Adam (Kingma, 2015) with a learning rate of 0.0001 and an effective batch size of 256, run for 150k, 300k, and 80k steps on ToyShape, SimObject, and RealHand, respectively.

Inference of Diffusion Models. For ancestral sampling (DDPM), the number of sampling steps is set to 1,000. For the ODE-based solvers, we evaluate one first-order and one second-order method: DPM-Solver-1, equivalent to DDIM (Song et al., 2021), and DPM-Solver-2 (Lu et al., 2022), respectively. Each solver is assessed at three standard step budgets: 25, 50, and 100 steps. On RealHand, the YOLO-based counting model is run with a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless stated otherwise, the number of generated samples equals the number of training samples in each evaluation, and all reported results are averaged over three independent random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports the counting hallucination rate (CHR), the non-counting failure rate (NCFR), and the total failure rate (TFR) as functions of the denoising configuration, namely the solver type, the ODE solver order, and the initial-noise setting, across ToyShape, SimObject, and RealHand. We summarize the salient observations below.

(1) Increasing the number of sampling steps reduces counting hallucinations on the synthetic datasets but amplifies them on RealHand. As reported in Table 1, under the standard ODE-solver step budgets (25, 50, and 100 steps), increasing the step count monotonically decreases the CHR on ToyShape and SimObject but increases it on RealHand, the sole exception being DPM-Solver-2 at 100 steps. We attribute this dichotomy to distributional complexity: the simpler, more structured synthetic distributions benefit from finer solver discretization, whereas the more complex RealHand

Table 1: Counting hallucination rate (CHR), defined as the number of counting hallucinations divided by the total number of generated samples, as a function of the solver configuration and initial-noise setting across ToyShape, SimObject, and RealHand. For RealHand we additionally report the non-counting failure rate (NCFR), i.e., the fraction of failures captured by the counting-ready indicator. The total failure rate (TFR) is the sum of CHR and NCFR. *Diffused* and *Normal* denote initialization from the ground-truth diffused noise and from standard Gaussian noise, respectively.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

distribution induces overfitting to local inconsistencies as the step count grows, thereby amplifying hallucinations.

(2) **Higher-order ODE solvers reduce non-counting failures but increase counting hallucinations on RealHand at the standard step budgets.** As reported in Table 1, comparing DPM-Solver-1 with DPM-Solver-2 on RealHand at the standard step budgets (e.g., 25 and 50 steps) exposes a consistent trade-off: the higher-order solver markedly lowers the NCFR while simultaneously raising the CHR. The same trend holds as the step count is increased for either solver. In aggregate, allocating additional computation (i.e., “better” sampling conditions) consistently improves perceptual quality, as evidenced by the lower NCFR, but does not enforce the factual correctness of the finger count.

(3) **DDPM substantially outperforms the ODE solvers on both counting hallucinations and non-counting failures.** Across all configurations, ancestral sampling (DDPM) attains the lowest CHR, NCFR, and TFR, establishing an empirical lower bound on the failure rates achievable in this setting. These results identify ancestral sampling as an effective strategy for minimizing hallucination.

(4) **A more informed initial-noise distribution reduces both counting hallucinations and non-counting failures.** As reported in Table 1, initialization from “diffused” noise (the ground-truth initial noise) yields lower CHR, NCFR, and TFR than initialization from “normal” noise (standard Gaussian noise). This gain is consistent with the diffused noise being better matched to the initial-noise distribution seen during training, indicating that a more informed initialization mitigates both counting hallucinations and non-counting failures.

(5) **Greater morphological complexity of the countable objects increases the incidence of counting hallucinations.** Comparing the CHR across ToyShape, SimObject, and RealHand shows that it increases monotonically with the morphological complexity of the target objects. This indicates that the fine-grained detail and complex spatial relationships characteristic of morphologically complex objects impede the model’s ability to preserve accurate object counts, yielding more frequent spurious additions and omissions.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Since counting hallucination is a form of factual hallucination, a natural question is: *does an improved perceptual metric, e.g., a lower FID, necessarily entail fewer counting hallucinations by virtue of a smaller distributional gap between the generated and training data?* We find that the answer is **negative in general**.

To characterize this relationship, we estimate the correlation between the CHR and perceptual metrics, and between the NCFR and the same metrics. We consider two widely used perceptual measures: a distribution-level metric, FID (Heusel et al., 2017), and an instance-level metric, MUSIQ (Ke et al., 2021). Correlation is quantified by three complementary coefficients: the Pearson correlation (Lee Rodgers & Nicewander, 1988) (linear dependence), the Spearman rank correlation (Spearman, 1904) (monotonic dependence), and the distance correlation (Székely et al., 2007) (general nonlinear dependence). The following evidence substantiates our conclusion.

Table 2: Correlation of the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) with the perceptual metrics FID and MUSIQ. The PTI-Hand results are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to synthesize 5050 samples, matching the size of RealHand. For PTI-Hand, FID is computed between the RealHand images and the model-generated samples. Here p denotes the p-value quantifying the statistical significance of the correlation r or ρ . The marker $\dagger$ indicates that the DDPM results are pooled with those of the various ODE solvers across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01						
Pearson † (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01	—	—						
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29						
Spearman † (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01	—	—						
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—	0.99	—						
Distance Correlation † (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—	—	—						

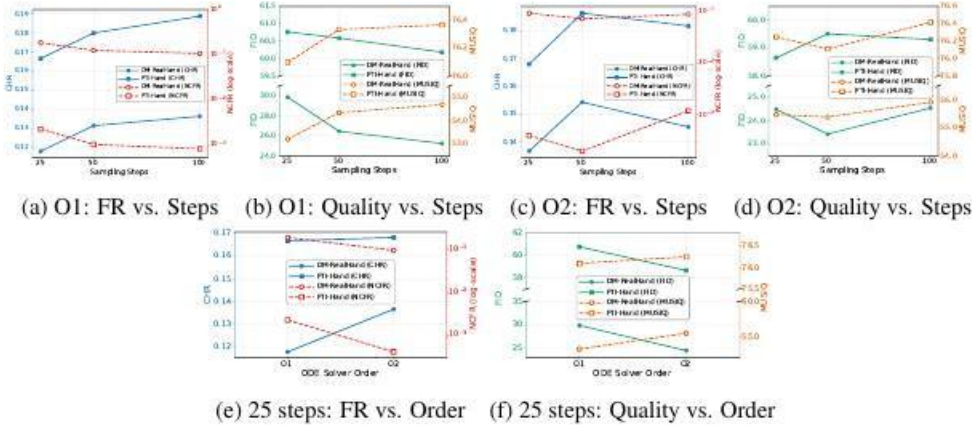


Figure 3: Trend consistency between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote the first- and second-order ODE solvers (DPM-Solver-1 / DPM-Solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). The failure-rate (FR) metrics are CHR and NCFR; the quality metrics are FID and MUSIQ. Under variation in both the number of sampling steps and the solver order, the two models exhibit highly consistent trends.

✎ **The correlation between CHR and the perceptual metrics is dataset- and solver-dependent rather than intrinsic.** As reported in the first and third rows of Table 2, the Pearson correlation between CHR and FID changes sign across datasets, being positive for SimObject, negative for RealHand, and again positive for PTI-Hand, and the corresponding Spearman correlation transitions from positive to negative to negative. The distance correlation likewise decreases from 0.82 (SimObject) to 0.74 (RealHand) to 0.47 (PTI-Hand), indicating that greater complexity of the generated content weakens the CHR–FID association. Pooling the DDPM results with those of the ODE solvers attenuates these correlations further.

✎ **The correlation between NCFR and the perceptual metrics is uniformly strong and positive.** In contrast to the weak CHR–metric correlations, the NCFR–metric correlations are strong and stable across all settings. For NCFR versus FID, for instance, every coefficient exceeds 0.88 with negligible p-values, denoting high statistical significance.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

We now assess counting hallucinations in a large-scale pretrained text-to-image model. Specifically, we evaluate Stable-Diffusion-3.5-Medium (Esser et al., 2024) using the counting-ready indicator and counting model introduced in Sec. 4.2.3; further details are given in Appendix A.8. As shown in Figure 3, the dependence of counting hallucinations on the solver type and the number of sampling steps in the large pretrained model closely matches the trends established by our controlled RealHand experiments. This agreement corroborates the findings of Sec. 5.2 and evidences the broader

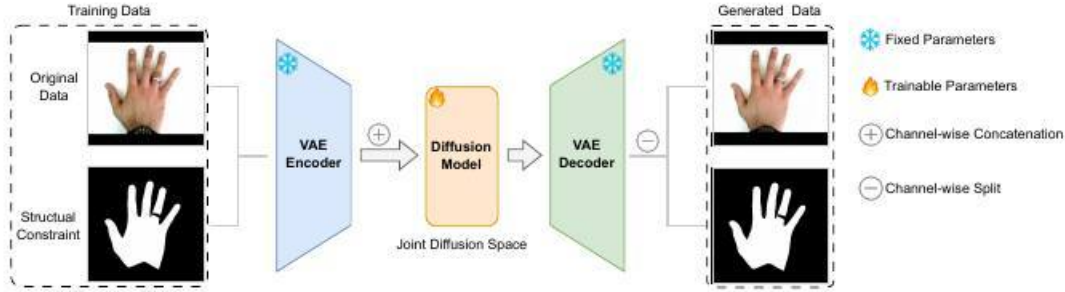


Figure 4: Schematic of the proposed joint-diffusion model (JDM). The model concatenates, along the channel dimension, the latent representations of the hand RGB image and its corresponding segmentation mask. This joint learning paradigm biases generation toward anatomically plausible hand poses and correct finger topology, thereby reducing both CHR and NCFR.

Table 3: Failure rates of the LDM baseline and the proposed JDM on the RealHand dataset across ODE solvers and sampling-step budgets; lower is better. Although JDM jointly generates the hand RGB image and its segmentation mask, FID is computed on the RGB outputs alone to preserve comparability with the baseline.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fr�chet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

applicability of the proposed evaluation protocol. The correlation results in Table 2 reinforce this conclusion: the CHR-metric correlations are again weak while the NCFR-metric correlations are again strong. Collectively, these results establish the robustness and generality of our findings and demonstrate that the proposed evaluation protocol transfers reliably to pretrained text-to-image models.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The lower counting hallucination rates observed for simpler countable objects (Observation 5 in Sec. 5.2) indicate that diffusion models capture counting-based factuality more reliably when the underlying visual structure is of low morphological complexity (e.g., masked objects). This motivates the following question: if a diffusion model were to denoise the general visual representation jointly with an explicit, low-complexity structural constraint that is intrinsically easier to model, such as an object mask encoding coarse topology and object counts, within a shared latent space, would this joint learning paradigm mitigate counting hallucinations? We term models that jointly denoise the general visual representation and the structural constraint **joint-diffusion models** (JDM).

Concretely, we concatenate, along the channel dimension, the latent representation of the hand mask with that of the corresponding RGB image, as shown in Fig. 4. We then fine-tune the UNet pretrained on CelebA-HQ (Karras et al., 2018), following the RealHand configuration of Sec. 5.1, on the RealHand images paired with their masks, which are extracted with SAM-2 (Ravi et al., 2024). To accommodate both modalities, the input and output channel dimensions of the denoising UNet are expanded to the combined latent dimensionality of the image and mask latents (i.e., $3 + 3 = 6$ channels).

As reported in Table 3, JDM attains lower failure rates than the vanilla LDM on both CHR and NCFR across all sampling conditions, evidencing its effectiveness at accurately generating structurally intricate objects that are susceptible to counting ambiguity. JDM, however, incurs a higher FID. We attribute this degradation to the additional denoising burden of jointly modeling two modalities under a small-scale training regime, which induces a trade-off of perceptual quality for improved structural fidelity. We further observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled fine-tuning settings. Additional details, results, and discussion are provided in Appendix A.9.

6 CONCLUSION

This work takes a first step toward the systematic quantification of hallucinations in diffusion models and provides new insight into improving factual generation. Restricting attention to counting hallucinations, we introduce the **CountHalluSet** suite together with a practical, reproducible methodology for measuring them. Through controlled quantification experiments, we establish that numerical refinements standard in diffusion sampling frequently fail to mitigate counting hallucinations in human-hand generation. Our correlation analyses further demonstrate that widely used perceptual metrics (e.g., FID and MUSIQ) do not consistently track the severity of counting hallucinations. Experiments on large-scale pretrained text-to-image models corroborate both the generality and the validity of these findings.

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A APPENDIX

A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

Consider the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the model reverse process $p_\theta(\mathbf{x}_{T:0})$. Their Kullback–Leibler divergence is

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

Invoking the Markov property, both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$ factorize as

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

Applying the chain rule for the KL divergence to this factorization yields:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

By the Data Processing Inequality (Cover & Thomas, 2012), marginalizing out a subset of the random variables cannot increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

Combining Eq. (A.4) with Eq. (A.5) yields the bound on the total accumulated distributional error:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

where equality holds if and only if $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$, i.e., precisely when the diffusion model exactly recovers the data distribution $q(\mathbf{x}_0)$. This upper bound on the overall KL divergence $\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0))$ between the true and model-generated data coincides, up to the data-entropy term $H(\mathbf{x}_0)$, with the variational bound of DDPM (Ho et al., 2020); consequently, minimizing the DDPM variational loss directly tightens this explicit bound on the discrepancy between the true and model-generated distributions.

Diffused-prior gap. The first error source is the mismatch between the distribution of the ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ and the sampling prior $p_\theta(\mathbf{x}_T)$. This mismatch is a consequence of the diffusion horizon T being finite in practice. It is quantified by a general distributional divergence:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence. The discrepancy may equivalently be evaluated under alternative distributional metrics, such as FID or the Wasserstein distance. Being

present at initialization, the diffused-prior gap propagates through the entire denoising trajectory and manifests as a persistent output error.

Noise model prediction error. The second error source is the noise prediction model. The estimate $\epsilon_\theta(\mathbf{x}_t, t)$ only approximates the true noise ϵ , with the approximation guaranteed in principle by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Denote the prediction error by $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$. By Eq. (5), any error in ϵ_θ induces a proportional deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Because each reverse step is conditioned on the mean produced by the preceding step, this bias accumulates along the denoising trajectory and contributes to the overall prediction error.

A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Starting from Eq. (7), we analyze a one-step numerical solver as a representative case. Let $\mathbf{x}^*(\tau)$ denote the exact trajectory satisfying the ODE under the true score function $h^*(\mathbf{x}(\tau), \tau)$, initialized at a ground-truth condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Correspondingly, let $\hat{\mathbf{x}}_k$ denote the trajectory produced by a numerical ODE solver at the discrete steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and let $\hat{h}$ denote the estimated model function. Our objective is to bound the total accumulated trajectory error of a given numerical trajectory $\hat{\mathbf{x}}_0, \dots, \hat{\mathbf{x}}_N$. Define the per-step trajectory error $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ and consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. Applying the variation-of-constants formula to the exact solution over one step (from τ_k down to τ_{k+1}) gives:

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator associated with the linear part $\frac{dz}{d\tau} = f(\tau)z$; note that the integration limits run backward in time.

Numerical path evolution. A one-step numerical solver approximates this solution, and its update deviates from the exact evolution under $\hat{h}$ initialized at $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ denote the exact solution at τ_{k+1} of the ODE driven by $\hat{h}$ and initialized at $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method incurs a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k, \quad (\text{A.11})$$

which gives:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtracting the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12) yields:

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. Unrolling this recurrence from $k = 0$ to $N - 1$, with the composition convention $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$ and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$, and

applying the recurrence repeatedly gives:

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$ gives:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated-initial-error term captures the deterministic contribution of a particular initial-error realization to the final trajectory error, whose statistical origin is the distributional diffused-prior gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

Following Eq. (2), the DDPM (Ho et al., 2020) forward process is

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

and the corresponding reverse process is

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to Eq. (A.16), whereas its deterministic reverse process (with $\eta = 0$) is

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share an identical forward (diffusion) process, DDIM is not obtained by merely deleting the stochastic term from the DDPM reverse (denoising) process, since the two methods assign different coefficients to the predicted-noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative samples from the three datasets used in this study. As reported in Table A1, each dataset is balanced across its categories, and Table A2 shows that the distribution over per-image object counts is comparable, confirming the absence of object-count bias across datasets. The datasets are listed in the same order in both tables; the category ordering in Table A1 is:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

Table A1: Per-category object counts for each dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Number of images stratified by per-image object count for each dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

We provide representative examples of counting hallucinations for the three datasets in Fig. A2. For ToyShape and SimObject, the counting hallucinations either contain two or more instances of at least one category, or contain no instances despite a well-formed background, as shown in Fig. A2(a) and Fig. A2(b). For RealHand, the counting-hallucinated samples are characterized by an incorrect finger count (e.g., 4 or 6 fingers), as shown in Fig. A2(c). Notably, the last two images in the first row of Fig. A2(c) appear to display five fingers; however, one finger is incomplete and visibly artifacted, yielding four valid fingers, so these samples are correctly classified as counting hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative YOLO predictions on the RealHand dataset. The detector localizes fingertips accurately across diverse hand poses and orientations, spanning both dorsal and palmar views. It nonetheless fails to flag certain anatomically implausible configurations, such as hands with a duplicated thumb (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers and no thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). Such counting-valid samples are still detected as five-finger hands and are therefore indistinguishable, by finger count alone, from anatomically valid ones.

Figure A3(b) shows counting-hallucinated samples for which YOLO predicts an incorrect fingertip count (e.g., 3, 4, 6, or 7 fingertips). Figure A3(c) further illustrates rare edge cases ($< 1\%$) in which the detector fails to localize fingertips with ill-defined boundaries, cracked fingers, or floating artifacts. These cases can be partially alleviated by tuning YOLO’s inference hyperparameters (e.g., the IoU and confidence thresholds), but doing so degrades detection on the majority of counting-valid samples. Because a fixed detection configuration is applied across all RealHand evaluations, these rare cases do not compromise the validity of our overall conclusions, notwithstanding that resolving them remains a difficult and low-reward endeavor.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows representative examples of the three classes of generated hand images: counting-valid, counting-hallucinated, and non-counting failure samples. As illustrated in Fig. 2, the RealHand evaluation procedure first applies the counting-ready indicator to separate counting-ready samples (counting-valid and counting-hallucinated) from images unsuitable for counting, the latter typically exhibiting severe visual artifacts such as distorted, cracked, partially visible, or floating fingers. The counting-ready samples are then partitioned into counting-valid and counting-hallucinated according to whether the detected number of valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

In the PTI-Hand experiments, we condition Stable-Diffusion-3.5-Medium on 10 distinct hand-generation prompts, enumerated below.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

The definition of hallucination presupposes that the model generates configurations never observed during training. This premise is difficult to verify for large-scale pretrained diffusion models, whose full training distribution is inaccessible. To render the quantification of counting hallucinations tractable in this setting, we adopt a conservative criterion: any generated hand image with six or more fingers is labeled counting-hallucinated. This criterion is well justified: whereas real images may depict fewer than five visible fingers due to occlusion, articulation, or viewpoint, real data are exceedingly unlikely to contain hands with six or more fingers. A finger count ≥ 6 therefore constitutes a reliable and empirically verifiable indicator of counting hallucination. Generated examples are provided in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

In the JDM experiments of Table 3, the denoising UNet is fine-tuned for 120k steps, versus 80k steps for the vanilla LDM. This larger training budget is required because extending the latent space from three to six channels induces a substantial mismatch with the pretrained LDM latent distribution. Additional optimization is therefore needed to close this distributional gap and allow the modeling capacity of JDM to be fully realized.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Table A3: Failure rates and image quality for fine-tuning SD-1.5 versus fine-tuning SD-1.5 under the joint-diffusion training paradigm on RealHand. The first row (baseline) reports fine-tuning Stable-Diffusion-1.5 on RealHand. The second row reports the variant in which all intermediate layers of the baseline are frozen and only the input and output layers with extended channel dimensions are fine-tuned. Lower is better.

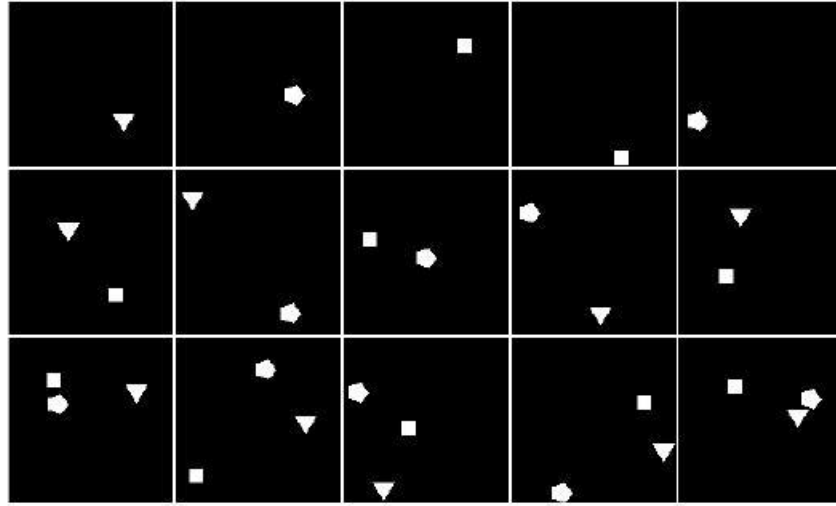
Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

Importantly, JDM should be regarded as a **novel training paradigm** for improving factual generation in diffusion models, not as a superior inference strategy. First, JDM extends only the input and

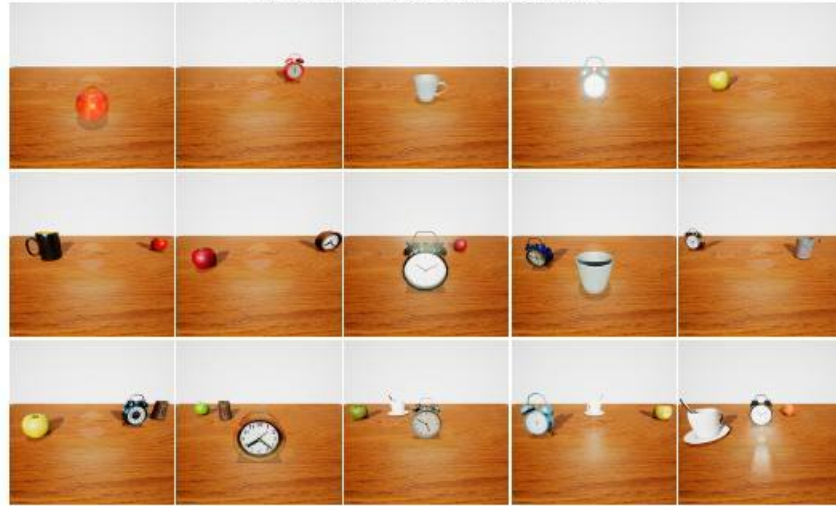
output channels of the UNet while leaving all intermediate-layer dimensions unchanged, and thus **introduces negligible inference-time overhead relative to the vanilla LDM**. Second, augmenting a pretrained text-to-image model with additional mask channels **perturbs the manifold geometry of the pretrained latent space and shifts the original denoising distribution**, rendering direct fine-tuning of such a model nontrivial.

Despite this difficulty, we still observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled fine-tuning conditions. Empirically, jointly aligning the extended latent space of JDM with the highly rigid latent geometry of large pretrained models is difficult given the limited scale of RealHand. To balance these competing constraints, we adopt a two-stage fine-tuning strategy: we first fine-tune a Stable-Diffusion-1.5 model on RealHand, then freeze all intermediate UNet layers and fine-tune only the input and output layers with extended channels. As reported in Table A3, even under this heavily constrained setup, the joint-diffusion paradigm consistently yields lower counting hallucination rates at 25 and 50 sampling steps for both solvers.

Overall, the evidence in this work indicates that JDM is a promising direction for enforcing factual consistency in image generation. Finally, we contend that **characterizing its behavior and performance under true large-scale training remains an important and compelling direction for future work**.



(a) Examples in the ToyShape dataset

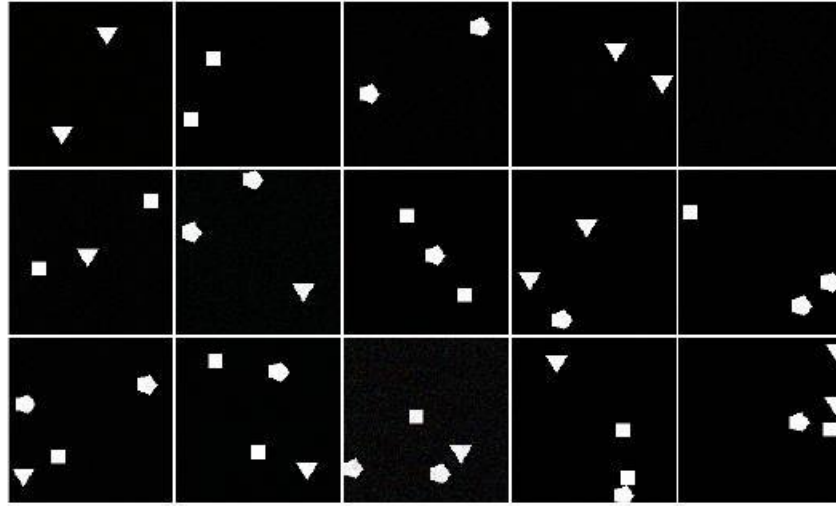


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

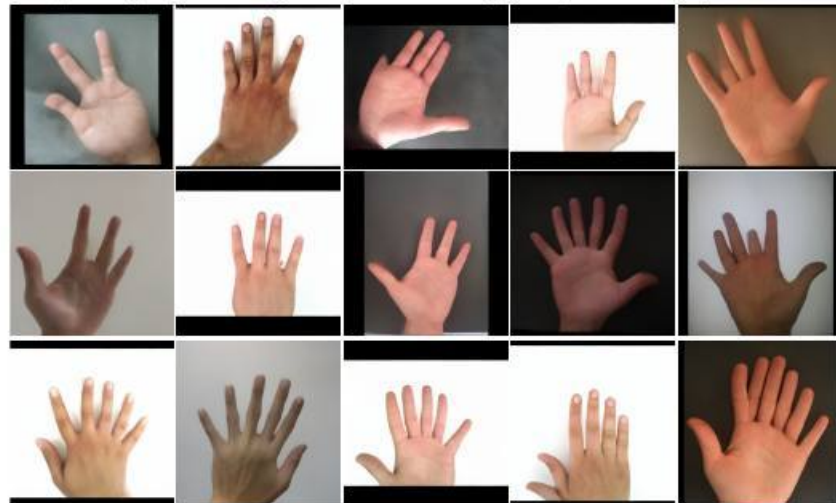
Figure A1: Representative samples from the datasets of the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Examples of counting hallucinations corresponding to the ToyShape dataset

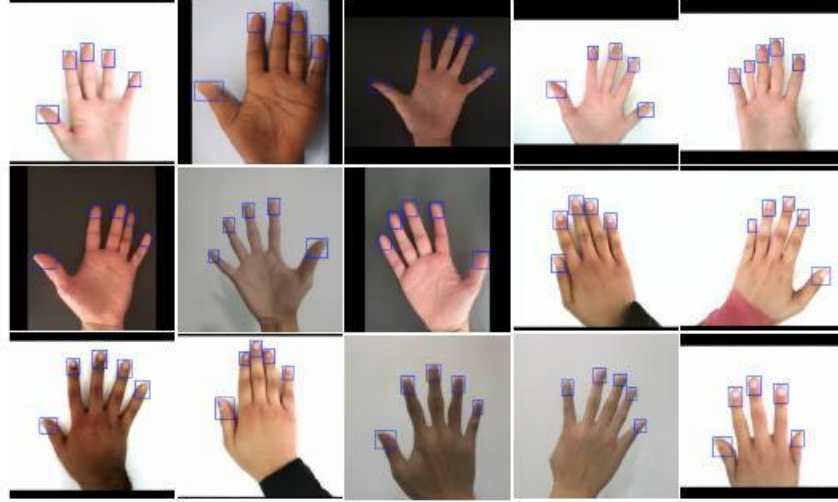


(b) Examples of counting hallucinations corresponding to the SimObject dataset

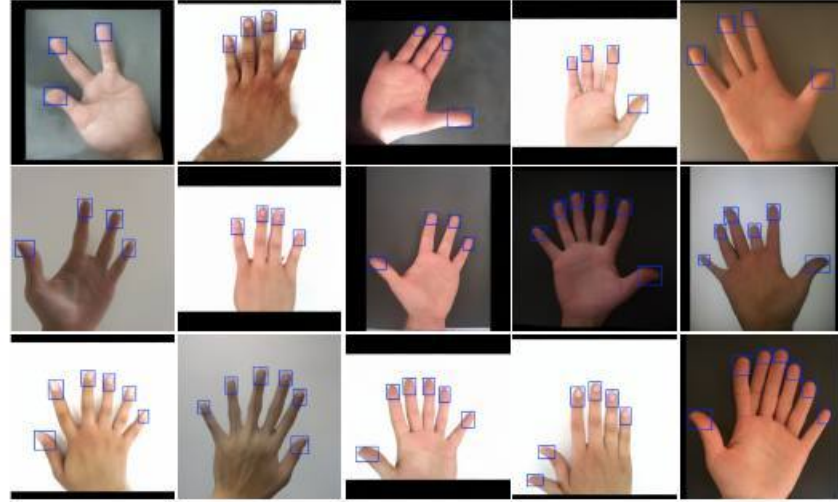


(c) Examples of counting hallucinations corresponding to the RealHand dataset

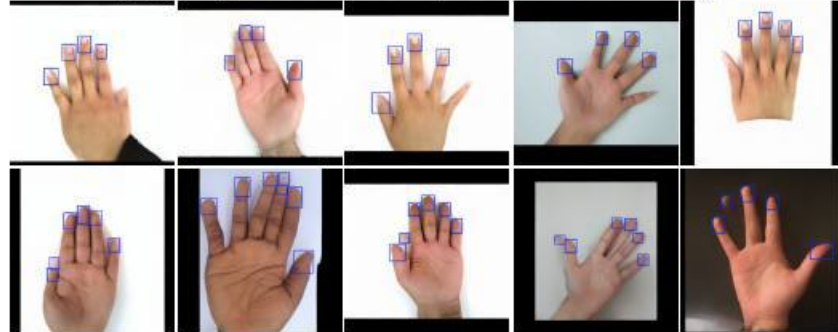
Figure A2: Representative counting hallucinations for the ToyShape, SimObject, and RealHand datasets.



(a) Counting-valid YOLO predictions with five fingers

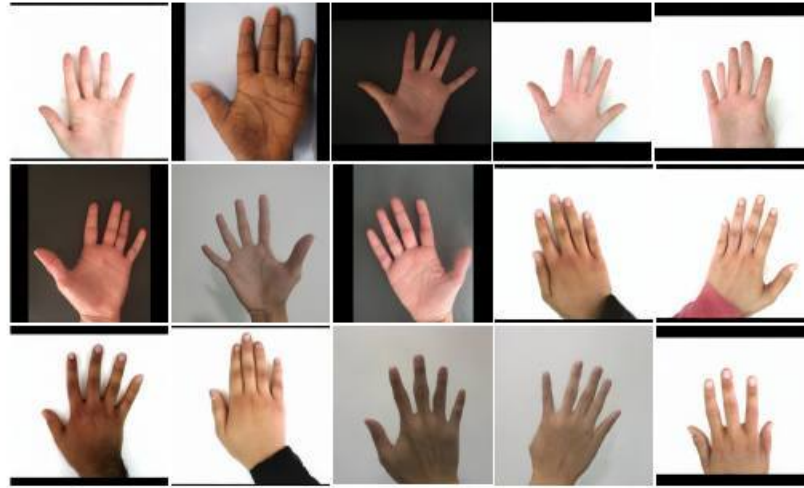


(b) Counting-hallucinated YOLO predictions with not five fingers

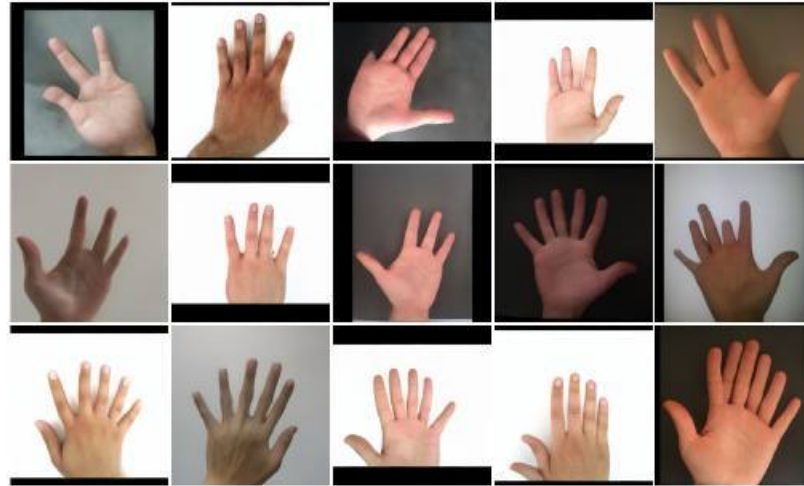


(c) Rare edge cases where YOLO predictions fail

Figure A3: YOLO predictions on generated RealHand images. (a) Counting-valid samples for which YOLO correctly predicts five fingertip bounding boxes (including cases with two thumbs). (b) Counting-hallucinated samples with an incorrect predicted fingertip count (e.g., 3, 4, 6, or 7 detected fingertips). (c) Rare edge cases that the counting-ready indicator fails to filter out and that YOLO handles poorly, such as blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: The three classes of generated hand images. (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with an incorrect number of valid fingers (e.g., 4 or 6). (c) Non-counting failures: images unsuitable for counting owing to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: The three classes of generated hand images produced by pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with more than five valid fingers (e.g., 6 or 7). (c) Non-counting failures: images unsuitable for counting owing to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.